CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION

Cambridge IGCSE™ & O Level Computer Science (0478 / 2210)

OFFICIAL MARKING SCHEME & STEP-BY-STEP WORKED SOLUTIONS (2023 – 2026)

TOPICS 1.1 (Number Systems) & 1.2 (Text, Sound, Images & Compression) **Maximum Mark: 65 Marks**

TOPIC 1.1: NUMBER SYSTEMS — WORKED SOLUTIONS

Question 1 Cambridge 2210/12/M/J/24 Q1 [Total: 6 marks]

(a) Binary to Denary: `01011010` = $64 + 16 + 8 + 2 =$ **90** [1 mark]
(b) Two's Complement of −45: [2 marks]
• Step 1: $+45$ in 8-bit binary = `00101101` [1]
• Step 2: Invert bits → `11010010` + 1 = **11010011** [1]
(c) Binary Addition & Overflow: [3 marks]
`01011010 (90) + 00110111 (55) = 10010001 (145)` [1]
• **No overflow error has occurred** [1]
• **Reason:** Result (145) fits within 8 bits (0–255), no carry out from the MSB [1].

Question 2 Cambridge 2210/11/M/J/24 Q1 [Total: 6 marks]

(a) Hexadecimal 3B Conversion: [2 marks]
• Binary: $3 = 0011,; ext{B} = 1011$ → **00111011** [1]
• Denary: $(3 imes 16) + 11 = 48 + 11 =$ **59** [1]
(b) Reasons for using Hex: [2 marks, 1 mark each]
• Shorter and more compact representation than binary (uses less screen/memory space).
• Much easier for humans to read, remember, and transcribe without errors.
• Fast 1-to-1 direct mapping with 4-bit binary nibbles.
(c) Applications: MAC addresses, IPv6 addresses, HTML/RGB colour codes, Memory dumps, Error codes [2 marks].

Question 3 Cambridge 2210/12/O/N/23 Q1 [Total: 5 marks]

(a) Logical Left Shift 2 places: [2 marks]
• `00110100` (52) shifted 2 places left → **11010000** [1]
• New Denary: $128 + 64 + 16 =$ **208** [1]
(b) Arithmetic Effect & Overflow: [3 marks]
• Each 1-place left shift multiplies the denary value by 2 (2 places multiplies by 4: $52 imes 4 = 208$) [1.5]
• If a `1` bit is shifted out of the MSB, an **overflow error** occurs, causing the number to lose precision and accuracy [1.5].

Question 4 Cambridge 0478/12/F/M/24 Q1 [Total: 5 marks]

(a) 8-bit Signed Range: Minimum = **−128** [1] • Maximum = **+127** [1]
(b) Two's Complement to Denary: [3 marks]
• Binary: `1 1 0 0 1 0 0 0`
• Place values: $-128 + 64 + 8 =$ **−56** [3 marks for correct working and answer]

TOPIC 1.2: TEXT, SOUND AND IMAGES — WORKED SOLUTIONS

Question 5 Cambridge 2210/12/M/J/23 Q1 [Total: 6 marks]

(a) ASCII vs Unicode: [3 marks]
• **ASCII:** Uses 7/8 bits ($128/256$ chars), supports English characters, numbers, control codes [1.5].
• **Unicode:** Uses 16/32 bits ($1 ext{M}+$ chars), supports all global languages, mathematical symbols, emojis [1.5].
(b) Text File Size Calculation: [3 marks]
• $2,048 imes 8 ext{ bits} = 16,384 ext{ bits} = 2,048 ext{ Bytes}$ [1]
• In KiB: $2,048 div 1,024 =$ **2 KiB** [2]

Question 6 Cambridge 2210/12/M/J/24 Q3 [Total: 7 marks]

(a) Sound Sampling Process: [3 marks]
• Sound waves are continuous analogue signals captured by a microphone [1].
• Analogue signal is sampled at regular time intervals using an ADC [1].
• Amplitude of each sample is measured and converted to a binary integer value [1].
(b) Definitions: [2 marks]
• **Sample Rate:** Number of sound samples captured per second in Hz [1].
• **Sample Resolution:** Number of bits allocated to store each sound sample [1].
(c) Effect of Increasing Sample Rate: File size increases [1] • Sound quality improves/becomes clearer [1].

Question 7 Cambridge 2210/11/O/N/23 Q2 [Total: 4 marks]

Sound File Size Calculation: [4 marks]
• $ ext{Bits} = 44,100 imes 16 imes 10 imes 1 = 7,056,000 ext{ bits}$ [1]
• $ ext{Bytes} = 7,056,000 div 8 = 882,000 ext{ Bytes}$ [1]
• In MB: $882,000 div 1,000,000 =$ **0.882 MB** (or $882,000 div 1,048,576 approx$ **0.841 MiB**) [2]

Question 8 Cambridge 0478/12/F/M/23 Q3 [Total: 7 marks]

(a) Image Definitions: [3 marks]
• **Pixel:** Smallest individual picture element in a bitmap image [1].
• **Resolution:** Total pixel dimensions ($ ext{Width} imes ext{Height}$) [1].
• **Colour Depth:** Number of bits used to represent the colour of each pixel [1].
(b) Image File Size: [4 marks]
• $ ext{Bits} = 1000 imes 800 imes 24 = 19,200,000 ext{ bits}$ [1]
• $ ext{Bytes} = 19,200,000 div 8 = 2,400,000 ext{ Bytes}$ [1]
• In MiB: $2,400,000 div 1,048,576 approx$ **2.29 MiB** (or $2.4 ext{ MB}$) [2]

Question 9 Cambridge 2210/12/O/N/24 Q2 [Total: 6 marks]

(a) Lossy vs Lossless: [4 marks]
• **Lossy:** Permanently discards redundant/imperceptible data; original cannot be recreated exactly (e.g. **JPEG, MP3**) [2].
• **Lossless:** Reduces file size without deleting data; original file reconstructed with zero loss (e.g. **PNG, ZIP, FLAC**) [2].
(b) Benefits of Compression: Faster transmission/download times [1] • Less bandwidth consumed and less storage required [1].

Question 10 Cambridge 2210/11/M/J/23 Q4 [Total: 4 marks]

(a) RLE Explanation: Lossless algorithm that identifies consecutive repeated values and encodes them as `[count, value]` pairs [2].
(b) RLE Solution: `W W W W W  B B B  W W  G G G G` → **5W 3B 2W 4G** (or `05 W 03 B 02 W 04 G`) [2 marks].